UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 20

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Now, assume that each measurement is noisy:

$$\text{measurement } i \in \{1, \dots, m\}: \quad b_i = \boldsymbol{a}_i^{\mathrm{T}} \boldsymbol{x}^* + \omega_i,$$

where ω_i is the unknown noise (e.g., sensor noise).

- Sensor noise is often Gaussian, and the best estimator for Gaussian (maximum likelihood) is least squares.
- So, to find $\omega_1, \dots, \omega_m$, we should minimize $\omega_1^2 + \cdots + \omega_m^2$ or $\|\boldsymbol{\omega}\|_2^2$.
- Also, to find a sparse $\boldsymbol{x}$, we should minimize $\|\boldsymbol{x}\|_0$.
- There are two objectives and we should convert them to a single one:

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{\omega} \in \mathbb{R}^m} \quad & \|\boldsymbol{\omega}\|_2^2 + \lambda \|\boldsymbol{x}\|_0, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} + \boldsymbol{\omega} = \boldsymbol{b}, \end{aligned}$$

  where λ is a user-defined positive constant balancing both objectives.

- As before, we consider a convex model:

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{\omega} \in \mathbb{R}^m} \quad & \|\boldsymbol{\omega}\|_2^2 + \lambda \|\boldsymbol{x}\|_1, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} + \boldsymbol{\omega} = \boldsymbol{b}. \end{aligned}$$

- This is a relaxation only when $\boldsymbol{x}$ is restricted to the box $\{\boldsymbol{y} \in \mathbb{R}^n \mid -1 \le y_i \le 1\}$.
- Since $\boldsymbol{\omega} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}$, we can eliminate $\boldsymbol{\omega}$ to arrive at

$$\min \quad \|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \lambda \|\boldsymbol{x}\|_1.$$

- This problem is called *Lasso*.

Lasso can be written as QP:

$$\begin{aligned} & \min_{\boldsymbol{x} \in \mathbb{R}^n} \quad \|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \lambda \|\boldsymbol{x}\|_1 \\ \iff & \min_{\boldsymbol{x}, \boldsymbol{t} \in \mathbb{R}^n} \quad \|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \lambda(t_1 + \cdots + t_n), \\ & \text{s.t.} \quad -t_i \le x_i \le t_i, \ i = 1, \dots, n \\ \iff & \min_{\boldsymbol{x}, \boldsymbol{t} \in \mathbb{R}^n} \quad \boldsymbol{x}^{\mathrm{T}}(\mathbf{A}^{\mathrm{T}}\mathbf{A})\boldsymbol{x} - 2\boldsymbol{b}^{\mathrm{T}}\mathbf{A}\boldsymbol{x} + \boldsymbol{b}^{\mathrm{T}}\boldsymbol{b} + \lambda(t_1 + \cdots + t_n), \\ & \text{s.t.} \quad -t_i \le x_i \le t_i, \ i = 1, \dots, n. \end{aligned}$$

The objective function is convex and quadratic in $(\boldsymbol{x}, \boldsymbol{t})$.

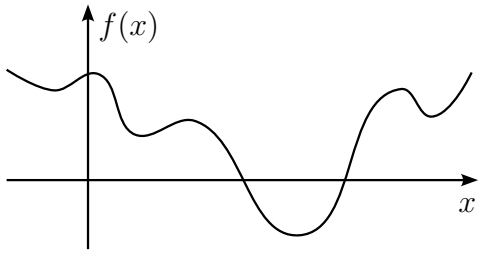
- Note: $\mathbf{A}^{\mathrm{T}}\mathbf{A}$ is PSD but not PD if $m < n$ (which is the case in compressed sensing).

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- So the Hessian of the objective is not PD. Hence, the objective may not be strictly convex and the solution of Lasso may not be unique.
- However, if $\mathbf{A}$ is random, then the solution is unique with probability 1 (i.e., those cases with multiple solutions are of measure 0 and rarely happen).
- Let $\hat{\boldsymbol{x}}$ denote the unique solution.
- There is a theorem (generalization of the informal theorem given in the previous Lecture Note) that says under some conditions on $\mathbf{A}, m$, if λ belongs to some good interval, then $\|\hat{\boldsymbol{x}} - \boldsymbol{x}^*\|$ is small and its value depends on how bad the noise $\boldsymbol{\omega}$ is.

Application: reconstruct a function from samples.

- Consider a function $\boldsymbol{f}\colon \mathbb{R}^n \to \mathbb{R}^k$ that is unknown and is to be learned.
- We can pick a number of points $\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}$ and obtain the value of $\boldsymbol{f}(\cdot)$ at those points.
- For many applications, calculating $\boldsymbol{f}(\boldsymbol{z}^{(i)})$ requires running a system or performing some simulation and it is costly.
- For example, $\boldsymbol{f}(\cdot)$ could represent the behavior of a robot with respect to some environment parameters, and obtaining $\boldsymbol{f}(\cdot)$ at each point means running the system in a certain environment.
- **Question:** How to learn $\boldsymbol{f}(\cdot)$ from as few samples as possible?
- Let's assume $n = k = 1$ (the general case is similar).



- We pick a suitable basis sets and approximate $f(x)$ as a combination of basis functions:

$$f(x) \approx \alpha_1 \sin x + \alpha_2 \sin 2x + \cdots + \alpha_p \sin px.$$

- Each continuous function can be written as such by selecting a large p, but many coefficients α_i's are close to zero (there are a few dominant frequencies in the function).

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- Define $b_i = f(z^{(i)})$, $i = 1, \dots, m$, then

$$b_i = \alpha_1 \sin z^{(i)} + \alpha_2 \sin 2z^{(i)} + \cdots + \alpha_p \sin pz^{(i)} + \omega_i$$

$$\Rightarrow \underbrace{\begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}}_{\boldsymbol{b}} = \underbrace{\begin{bmatrix} \sin z^{(1)} & \sin 2z^{(1)} & \cdots & \sin pz^{(1)} \\ \vdots & \vdots & & \vdots \\ \sin z^{(m)} & \sin 2z^{(m)} & \cdots & \sin pz^{(m)} \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_p \end{bmatrix} + \begin{bmatrix} \omega_1 \\ \vdots \\ \omega_m \end{bmatrix}.$$

  Here the noise ω_i takes care of the approximation error.

- Find $\alpha_1, \dots, \alpha_p, \omega_1, \dots, \omega_m$ satisfying the above equation so that most of α_i's are zero and ω_i's are small.

- Note: since we want to have as few samples as possible, m should be small and so $[\alpha_1, \dots, \alpha_p]^{\mathrm{T}}$ should be sparse to make sure equations are not too underdetermined:

$$\min_{\alpha_1,\dots,\alpha_p} \underbrace{\left\| \mathbf{A} \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_p \end{bmatrix} - \boldsymbol{b} \right\|_2^2}_{\substack{\omega_i\text{'s may not be Gaussian but} \\ \text{least-square ensures that these} \\ \text{approximation errors are small}}} + \lambda \left\| \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_p \end{bmatrix} \right\|_1.$$

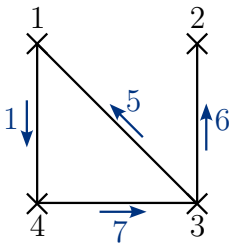
- The required number of samples m depends on the sparsity of the solution or how many important frequencies exist in $f(x)$.

Application in cyber attack detections in networks (power networks).

- Network: a group of nodes/vertices that are connected to each other via a set of edges/links.

- Each edge is associated with some flow.

- This is called a *flow network*.

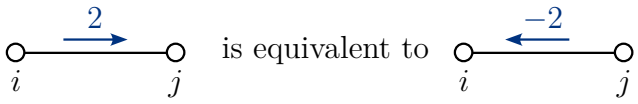


- Examples of flow:
  - Data (communication networks).
  - Water (water networks).

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- Electricity (power networks).
- Cars (transportation networks).

- Direction of flow over a line:

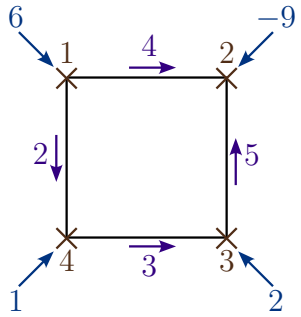


Flow from node i to node j is equivalent to negative flow from node j to node i.

- Define P_{ij} as the flow from node i to node j ($P_{ji} = -P_{ij}$).
- Define P_i as the injection at node i.
- Conservation of flow:

$$P_i = \sum_{(i,j)\in\text{edge}} P_{ij}, \quad i = 1, \ldots, n.$$

- Example:

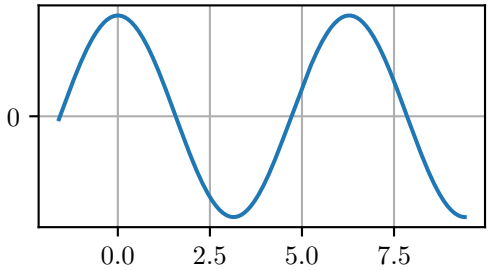


Node 1: $P_1 = P_{12} + P_{14} = 6 > 0 \Rightarrow$ produces commodity (power, water, etc.).

Node 2: $P_2 = P_{21} + P_{23} = -9 < 0 \Rightarrow$ consumes commodity.

Let's focus on power networks.

- The U.S. national grid is the largest machine on earth with millions of devices and over 200,000 miles of high-voltage transmission lines. It has over 80,000 nodes:
  - Suppliers (generators) produce electricity ($P_i > 0$);
  - Consumers (loads) use electricity ($P_i < 0$).
- Each node of the network has a voltage signal that is sinusoidal:



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- Frequency: 60 Hz in U.S.; 50 Hz in Europe.
- Magnitude: known (110/220 volts, etc.).
- Parameter to learn: phase.

  Voltage signal as a function of time:

  $$\text{magnitude} \times \cos(\omega t + \theta),$$

  where $\omega = 2\pi \times$ frequency, t is the time, and $\theta \in [-\pi, \pi]$ is the phase.
- Let $\theta_1^*, \ldots, \theta_n^*$ denote the unknown phases at nodes $1, \ldots, n$.
- Each edge/line (i, j) has a known parameter called reactance $Z_{ij} \in \mathbb{R}$.
- Flow $P_{ij} = (\theta_i^* - \theta_j^*)/Z_{ij}$ (this is a linearized model used in the industry).
- Note that if all θ_i^*'s are shifted by a constant r, all powers remain the same, since $\theta_i^* - \theta_j^*$ does not change. So, in power systems, we assume $\theta_1^* = 0$ and it serves as a reference.
- There are sensors installed at certain nodes and certain lines to measure injections and flows.
- The data collected from sensors is transmitted to a control center to process and learn $\theta_1^*, \ldots, \theta_n^*$.
- If a hacker tampers with the data, the resulting phases will be incorrect, causing the operator to implement a decision based on physically impossible values, which could trigger a large-scale blackout. For example, the 2015 cyberattack on Ukraine's power grid is a major security concern for the U.S. as well.
- Define $\hat{P}_i$ as the data received by control center for node $i \in \hat{\mathcal{N}}$, where $\hat{\mathcal{N}}$ is the set of nodes with sensors; define $\hat{P}_{ij}$ as the data received by control center for line $(i, j) \in \hat{\mathcal{E}}$, where $\hat{\mathcal{E}}$, where $\hat{\mathcal{E}}$ is the set of lines/edges with sensors. We have

  $$\hat{P}_i = \sum_{j \in \mathcal{N}_i} P_{ij} + \omega_i^* + v_i^*,$$

  where $\mathcal{N}_i$ is the set of neighboring nodes at node i, ω_i^* is the sensor noise, and v_i^* is the value added by the cyber attacker;

  $$\hat{P}_{ij} = P_{ij} + \omega_{ij}^* + v_{ij}^*,$$

  where ω_{ij}^* is the sensor noise and v_{ij}^* is the value added by the attacker.

  Note: only a few of v_i^*'s and v_{ij}^*'s are non-zero, since if an attacker tampers a lot of data, the operator would detect it and takes action to nullify the effect. Stealth attack requires a few attacks with a global impact, meaning that most (or all) phases learned by the control center become incorrect due to a few attacks.

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- Lasso:

$$\begin{aligned}\min_{\boldsymbol{\omega},\boldsymbol{v},\boldsymbol{\theta}} \quad & \|\boldsymbol{\omega}\|_2^2 + \lambda\|\boldsymbol{v}\|_1, \\ \text{s.t.} \quad & \hat{P}_i = \sum_{j\in\mathcal{N}_i} \frac{\theta_i - \theta_j}{Z_{ij}} + \omega_i + v_i, \ \forall i \in \hat{\mathcal{N}}, \\ & \hat{P}_{ij} = \frac{\theta_i - \theta_j}{Z_{ij}} + \omega_{ij} + v_{ij}, \ \forall (i,j) \in \hat{\mathcal{E}}, \\ & \theta_1 = 0, \qquad \text{(references for phases)}\end{aligned}$$

  where $\boldsymbol{\theta}$ is the vector of θ_i's, $\boldsymbol{\omega}$ is the vector of ω_i's and ω_{ij}'s, and $\boldsymbol{v}$ is the vector of v_i's and v_{ij}'s.

- Let $\hat{\theta}_1, \ldots, \hat{\theta}_n$ denote the optimal values of phases, and $\hat{\boldsymbol{v}}$ denote the optimal value of $\boldsymbol{v}$.

- Using the compressed sensing idea, it can be shown that if the number of attacks is small, then $\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|$ is small (the value depends on how big the noises ω_i^*'s and ω_{ij}^*'s are), $\hat{v}_i = 0 \Leftrightarrow v_i^* = 0$, $\hat{v}_{ij} = 0 \Leftrightarrow v_{ij}^* = 0$, and $\hat{\boldsymbol{v}}$ correctly identifies the locations of the attack.

  Therefore, by checking which elements of $\hat{\boldsymbol{v}}$ are non-zero, we discover whether and where there was a cyber attack.

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